LIMITS AND COLIMITS OF ABELIAN PRESHEAVES EXIST AND ARE COMPUTED SECTIONWISE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof's closing verification of the universal property is expressed by the author as straightforward; the construction and the comparison maps, which are the substance, are written out in the span.

Allowed standard prerequisites (editorial).

- Limits and colimits in the category of abelian groups. Stacks Project, Algebra, section on limits and colimits of modules (existence in abelian groups). Provides the objectwise limits and colimits used to define the candidate presheaves.

Lemma 0.1. *Let $\mathcal{C}$ be a category.*

(1) *All limits and colimits exist in $PAb(\mathcal{C})$.*
(2) *All limits and colimits commute with taking sections over objects of $\mathcal{C}$.*

Proof. Let $\mathcal{I} \to PAb(\mathcal{C})$, $i \mapsto \mathcal{F}_i$ be a diagram. We can simply define abelian presheaves L and C by the rules

$$L : U \longmapsto \lim_i \mathcal{F}_i(U)$$

and

$$C : U \longmapsto \operatorname{colim}_i \mathcal{F}_i(U).$$

It is clear that there are maps of abelian presheaves $L \to \mathcal{F}_i$ and $\mathcal{F}_i \to C$, by using the corresponding maps on groups of sections over each U. It is straightforward to check that L and C endowed with these maps are the limit and colimit of the diagram in $PAb(\mathcal{C})$. This proves (1) and (2). Details omitted. □

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